CBSE AI Practical 2026-27

Pradhumya Singh Yadav | Dhairya Chaudhary | Aryan Chauhan
Jagran Public School Noida

Class: XIIth

Problem

Diagnostic Gap: The inability of non-specialist health workers to accurately distinguish between complex respiratory sounds leads to high misdiagnosis rates.

Inaccessible Technology: Advanced digital stethoscopes cost upwards of ₹35,000, making AI diagnostic tools unaffordable for local clinics and rural PHCs.

The "Trial-and-Error" Risk: Local doctors often prescribe generic medication for serious symptoms, wasting critical time while underlying infections like Pneumonia worsen.

Critical Referral Delays: lack of early, objective detection for high-risk conditions results in late-stage hospitalizations and increased mortality in respiratory cases.

The Opportunity

Real-time AI classification allows local doctors to identify high-risk pathologies instantly, ensuring life-saving treatment begins days earlier.

Using low-cost, off-the-shelf hardware and open-source AI, this solution can be rapidly deployed across global regions with limited medical resources.

Converting analog breath sounds into digital data enables the creation of a longitudinal health record for better patient monitoring and follow-up.

By reducing the cost of digital stethoscopes by 80%, we can equip every ASHA worker and rural clinic with specialist-level screening tools.

Our Idea

1. VITALIS uses an embedded neural network to instantly translate raw lung sounds into clinical labels like "Normal," "Wheeze," or "Crackle."

2. By utilizing a high-fidelity I2S MEMS microphone, the device captures crystal-clear respiratory data without the electrical noise of traditional analog sensors

3. All AI inference is performed locally on the XIAO C6 microcontroller, ensuring diagnostic results are available in under 6ms without needing an internet connection.

4. The system provides a standardized "second opinion" to local healthcare workers, reducing human error and ensuring high-risk patients are referred to hospitals immediately.

SDGS

SDG 3: Good Health and Well-being

- The Problem: Respiratory diseases are leading causes of mortality in rural areas due to misdiagnosis.
- The VITALIS Solution: Providing an automated "second opinion" for lung disease for early detection and timely intervention.

SDG 9: Industry, Innovation, and Infrastructure

- The Problem: Commercial digital stethoscopes are prohibitively expensive
- The VITALIS Solution: Utilizes "Edge Computing" via neural networks to provide diagnostic infrastructure with zero INTERNET dependency.

Benefits

Elimination of Diagnostic Subjectivity

Replaces inconsistent human hearing with a standardized AI model, ensuring every patient receives an objective and accurate lung assessment.

Prevention of Critical Complications

Enables early detection of high-risk pathologies, allowing for timely hospital referrals and reducing the risk of minor infections turning into emergencies.

Cost-Effective Healthcare Infrastructure

Utilizes affordable, off-the-shelf hardware to provide professional-grade digital stethoscopy at a fraction of the cost of current market alternatives.

Enhanced Rural Medical Reach

Empowers frontline health workers and ASHA volunteers to perform complex diagnostic screenings in remote areas without specialist presence.

Empathy Map (Dr.)

Says

- "I need diagnostic certainty, not subjective acoustic noise from an analog tube."
- "A ₹40,000 digital stethoscope is completely out of my clinic's budget."

Thinks

- "If I miss fluid buildup today, this patient risks severe pneumonia by tomorrow."
- "I am forced to rely on auditory guesswork instead of objective medical data.

Does

- Prescribes broad-spectrum antibiotics preemptively to cover diagnostic blind spots.
- Delays critical hospital referrals until physical symptoms become severe.

Feels

- Paralyzed by the clinical liability of a misdiagnosis.
- Handicapped by primitive tools in a high-stakes environment.

Empathy Map (Patient)

Says

- "I cannot afford to lose a day's wage traveling 50km to the district hospital for a simple cough."
- "The local doctor gave me syrup, but my child is still struggling to breathe."
- "I don't have the money for expensive city tests right now."

Thinks

- "Is this just a seasonal cold, or something that will hospitalize me and bankrupt my family?"
- "Did the local doctor actually hear what is wrong, or is he just guessing?"
- "If I wait a few more days, maybe it will just go away on its own."

Does

- Ignores early-stage symptoms due to the logistical and financial friction of travel.
- Spends out-of-pocket money on generic, ineffective medications that mask the real disease.
- Arrives at the emergency room only when the respiratory condition is critical and life-threatening.

Feels

- Financially anxious about the hidden costs of healthcare and travel.
- Vulnerable to misdiagnosis at the local level.
- Terrified when their child's symptoms do not improve after basic treatment.

5W1H

3. 5W1H ANALYSIS

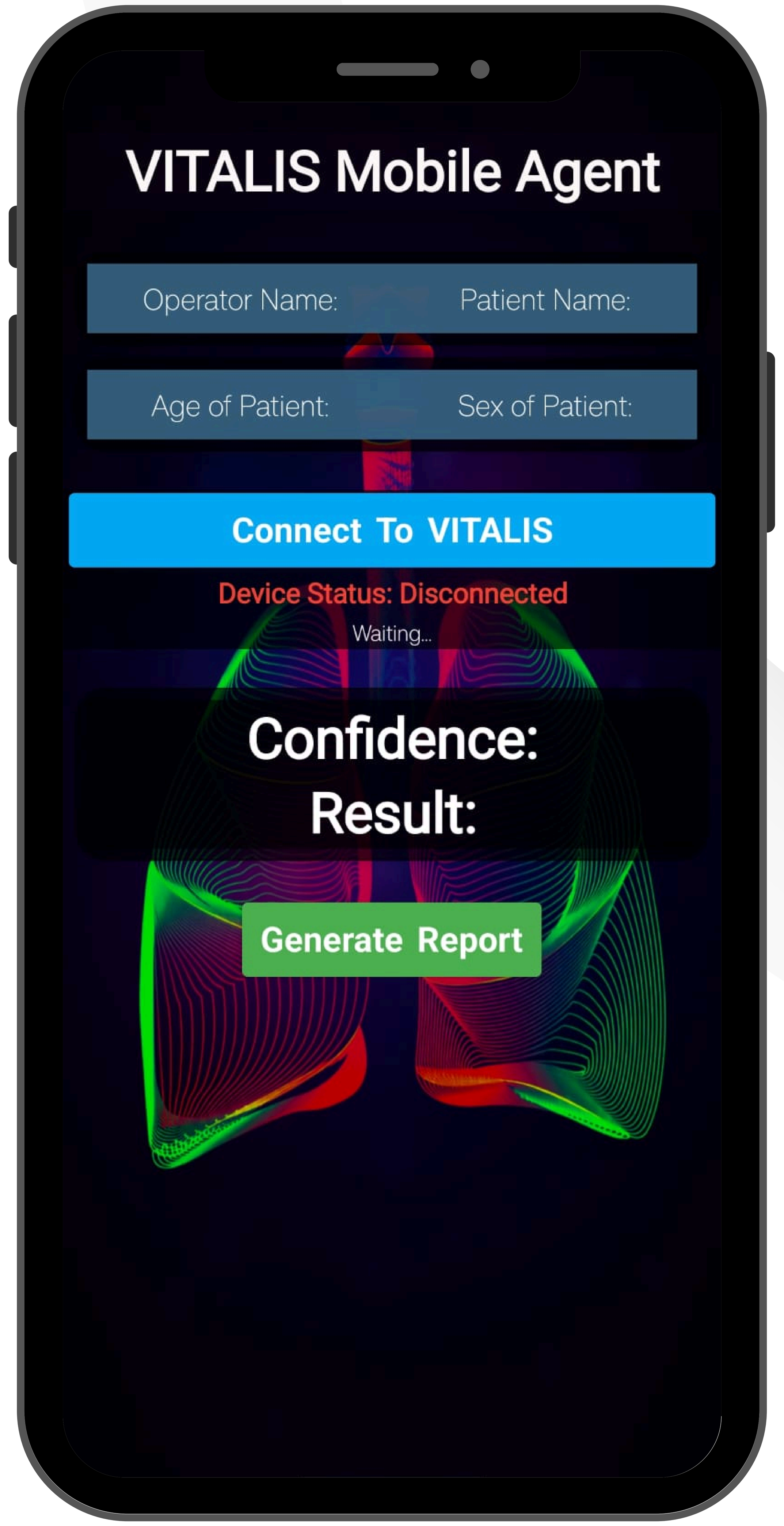
Question	Specification
Who?	ASHA Workers, NGO volunteers, and PHC medical officers across rural India.
What?	An Edge-AI hardware attachment that digitizes and analyzes lung acoustics in real-time.
Where?	Focused on Primary Health Centers (PHCs) and Community Health Centers (CHCs) in underserved districts.
When?	During the first point of contact in medical camps or rural check-ups.
Why?	To reduce infant mortality rates (IMR) caused by late-detected Pneumonia and manage high COPD prevalence.
How?	Through Digital Signal Processing (DSP) and int8 quantized Neural Networks running on the ESP32-C6.

Flow Chart

Start
Input: INMP441 Captures Audio Data
Processing
Process: I2S Digital Conversion
Process: DSP Feature Extraction
Is it Normal?
Yes
No
Is it Wheeze?
Yes
No
Is it Crackles?
Yes
No
Is it Rhonchi?
Yes
No
Output: Send 'Normal' to App
Output: Send 'Wheeze' to App
Output: Send 'Crackles' to App
Output: Send 'Rhonchi' to App
Output: Send 'Unknown' to App
Stop

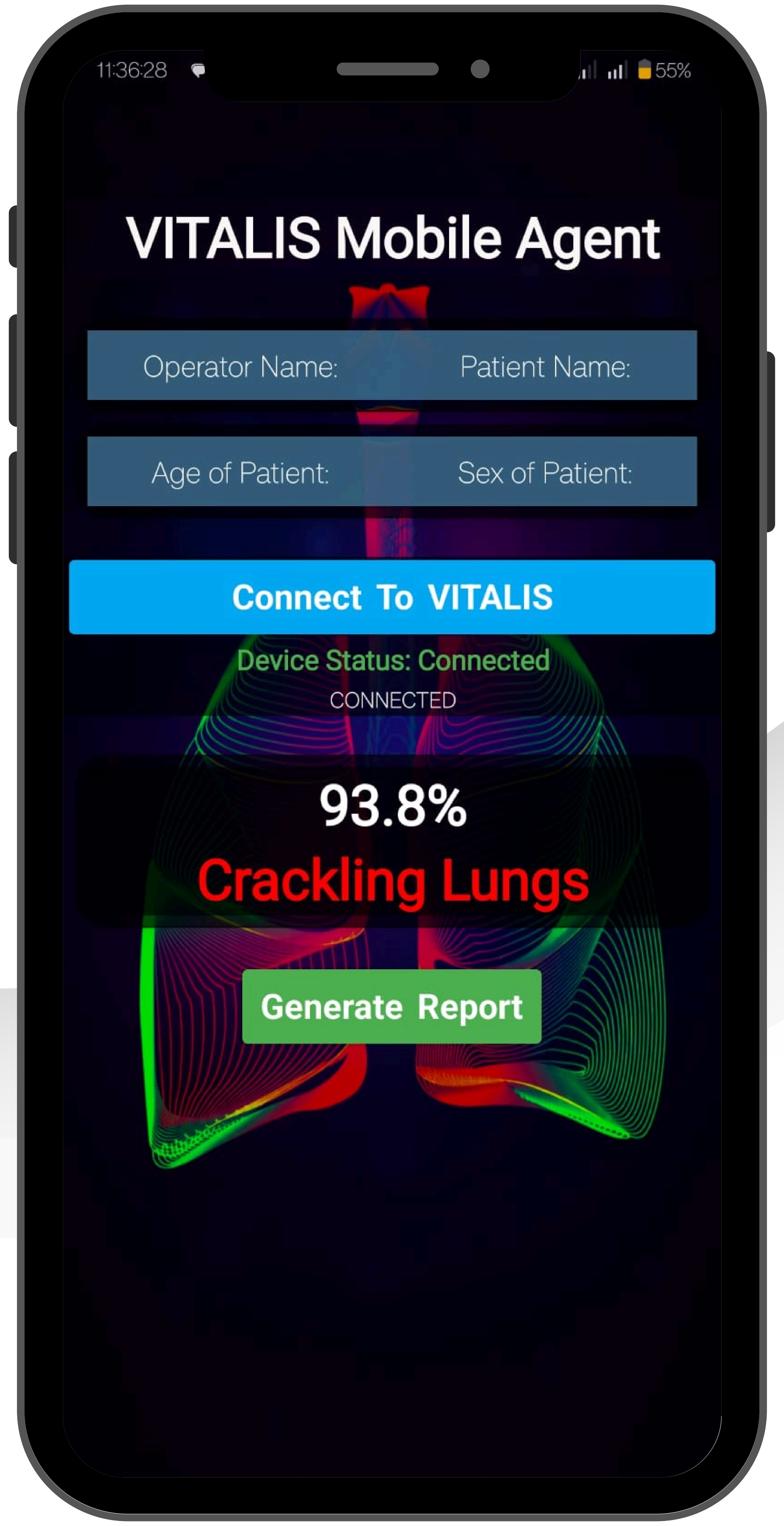
Prototype Image

Mobile APP Screenshot

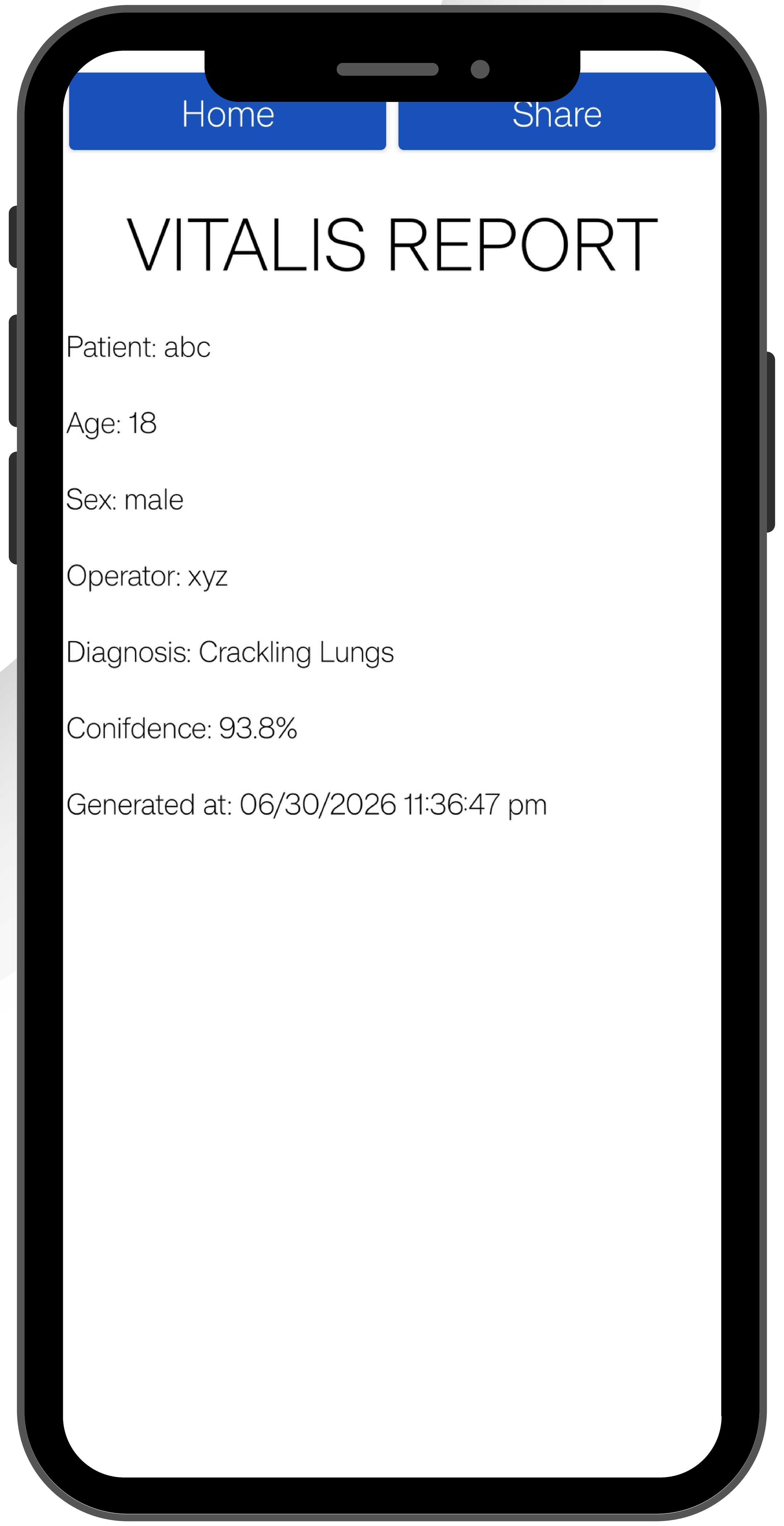


Open The App

Connect To VITALIS through Bluetooth



Press the Button On VITALIS to get Results



Generate the Report


VITALIS
THANK
YOU